Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

What is the angular velocity (ω) of an electron occupying second orbit of Lii^{2+} ion?

(A) $\frac{8\pi^3 \mathrm{me}^4}{\mathrm{h}^3}\ \mathrm{K}^2$

(B) $\frac{8\pi^3 me^4}{9h^3} K^2$

(C) $\frac{64}{9} \times \frac{\pi^3 \mathrm{me}^4}{\mathrm{h}^3}\ \mathrm{K}^2$

(D) $\frac{9\pi^3 \mathrm{me}^4}{\mathrm{h}^3}\ \mathrm{K}^2$

Q.2

The energy of an electron in first Bohr orbit of $\mathrm{H}-$ atom is $-13.6\mathrm{eV}$. The energy value of electron in the excited state of Li^{2+} is:

(A) $-27.2\mathrm{eV}$

(B) $30.6\mathrm{eV}$

(C) $-30.6\mathrm{eV}$

(D) $27.2\mathrm{eV}$

Q.3

The Bohr's energy equation for H atom reveals that the energy level of a shell is given by $\mathrm{E} = -13.58/\mathrm{n}^2\mathrm{eV}$. The smallest amount that an H atom will absorb if in ground state is

(A) $1.0\mathrm{eV}$

(B) $3.39\mathrm{eV}$

(C) $6.79\mathrm{eV}$

(D) $10.19\mathrm{eV}$

Q.4

If m and e are the mass and charge of the revolving electron in the orbit of radius r for hydrogen atom, the total energy of the revolving electron will be:

(A) $\frac{1}{2}\frac{ke^2}{r}$

(B) $-\frac{ke^2}{r}$

@IITJEE_Advanced

(C) $\frac{kme^2}{r}$

(D) $-\frac{1}{2}\frac{ke^2}{r}$

Q.5

An electron, e_1 is moving in the fifth stationary state, and another electron e_2 is moving in the fourth stationary state. The radius of orbit of electron, e_1 is five times the radius of orbit of electron, e_2 calculate the ratio of velocity of electron e_1 (v_1) to the velocity of electron e_2 (v_2)

(A) $5:1$

(B) $4:1$

(C) $1:5$

(D) $1:4$

Q.6

In a hydrogen atom, if energy of an electron $1n$ ground state is $13.6.ev$, then that in the 2nd excited state is

(A) $1.51eV$

(B) $3.4eV$

(C) $6.04eV$

(D) $13.6eV.$

Q.7

The Li^{2+} ion is moving in the third stationary state, and its linear momentum is $7.3 \times 10^{-34}kgms^{-1}$ Angular momentum is.

(A) $1.158 \times 10^{-45}\ kg\ m^2\ s^{-1}$

(B) $11.58 \times 10^{-44}\ kg\ m^2\ s^{-1}$

(C) $11.58 \times 10^{-47}\ kg\ m^2\ s^{-1}$

(D) $12 \times 10^{-45}\ kg\ m^2\ s^{-1}$

Q.8

An electron in an atom jumps in such a way that its kinetic energy changes from x to $\frac{x}{4}$. The change in potential energy will be:

(A) $+\frac{3}{2}x$

(B) $-\frac{3}{8}x$

(C) $+\frac{3}{4}x$

(D) $-\frac{3}{4}x$

@IITJEE_Advanced

Q.9

What atomic number of an element "X" would have to become so that the 4 th orbit around X would fit inside the I Bohr orbit of H atom?

(A) 3

(B) 4

(C) 16

(D) 25

Q.10

If in Bohr's model, for unielectronic atom, time period of revolution is represented as $\mathrm{T}_{\mathrm{n,z}}$ where n represents shell no. and Z represents atomic number then the value of $\mathrm{T}_{1,2} : \mathrm{T}_{2,1}$ will be :

(A) $8 : 1$

(B) $1 : 8$

(C) $1 : 1$

(D) $1 : 32$

Q.11

The energy of e^{-} in first orbit of He^{+} is $-871.6 \times 10^{-20}\ \mathrm{J}$. The energy of e^{-} in first orbit of H is:

(A) -871.6×10^{-20}

(B) $-435.8 \times 10^{-20}\ \mathrm{J}$

(C) $-217.9 \times 10^{-20}\ \mathrm{J}$

(D) -108.9×10^{-20}

Q.12

If radius of second stationary orbit (in Bohr's atom) is R then radius of third orbit will be:

(A) $\mathrm{R}/3$

(B) $9\mathrm{R}$

(C) $\mathrm{R}/9$

(D) $2.25\mathrm{R}$

Q.13

Bohr's atomic model can explain

(A) the spectrum of hydrogen atom only

@IITJEE_Advanced

(B) the spectrum of an atom or ion containing one electron only

(C) the spectrum of hydrogen molecule

(D) the solar spectrum

Q.14

Ionisation energy of He^{+} is 19.6×10^{-18} J atom $^{-1}$. The energy of the first stationary state $(n=1)$ of Li^{2+} is

(A) 4.41×10^{-16} J atom $^{-1}$

(B) -4.41×10^{-17} J atom $^{-1}$

(C) -2.2×10^{-15} J atom $^{-1}$

(D) 8.82×10^{-17} J atom $^{-1}$

Q.15

The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is $\left[a_0\right.$ is Bohr radius] :

(A) $\frac{h^2}{4 \pi^2 m a_0^2}$

(B) $\frac{h^2}{16 \pi^2 m a_0^2}$

(C) $\frac{h^2}{32 \pi^2 m a_0^2}$

(D) $\frac{h^2}{64 \pi^2 m a_0^2}$

Q.16

The potential energy of an electron present in the ground state of Li^{2+} ion is

(A) $-\frac{3 \mathrm{e}^2}{4 \Pi \in_0 \mathrm{r}^2}$

(B) $-\frac{3 \mathrm{e}}{4 \Pi \in_0 \mathrm{r}^2}$

(C) $\frac{3 \mathrm{e}^2}{4 \Pi \in_0 \mathrm{r}^2}$

(D) $-\frac{2 e^2}{4 \pi \varepsilon_0 r^2}$

Q.17

The ratio of kinetic energy and total energy of an electron in Bohr orbit of hydrogen-like species is

(A) $1/2$

(B) $1/4$

(C) -1

@IITJEE_Advanced

(D) $-1/2$

Q.18

The speed of an electron in the first Bohr orbit is $7.27 \times 10^5\ \mathrm{m\ s^{-1}}$. The speed of electron in the $n = 2$ level of $\mathrm{Li^{2+}}$ ion will be

(A) $7.27 \times 10^5\ \mathrm{m\ s^{-1}}$

(B) $1.09 \times 10^6\ \mathrm{m\ s^{-1}}$

(C) $1.45 \times 10^6\ \mathrm{m\ s^{-1}}$

(D) $1.82 \times 10^6\ \mathrm{m\ s^{-1}}$

Q.19

Which of the following statements regarding Bohr theory of hydrogen atom is not correct?

(A) Kinetic energy of an electron in an orbit is half of magnitude of its potential energy.

(B) Kinetic energy of an electron in an orbit is equal to the magnitude of its potential energy.

(C) Total energy of an electron is negative of its kinetic energy.

(D) Potential energy of an electron in an orbit is equal to $-mv^2$.

Q.20

The splitting of spectral lines of hydrogen atom in the presence of magnetic field is known as

(A) Stark effect

(B) Zeeman effect

(C) Bohr effect

(D) Anti-Zeeman effect

Answers & Solutions

Q.1 Answer:

$\frac{9\pi^3 \mathrm{me}^4}{\mathrm{h}^3}\ \mathrm{K}^2$

Solution:

$$\mathrm{v_n} = \mathrm{r_n}\omega \text{ where } \mathrm{r_n} = \frac{\mathrm{n}^2\,\mathrm{h}^2}{4\pi^2 \mathrm{me}^2 \mathrm{Z}\cdot\mathrm{K}}$$

$$\text{and } \mathrm{v_n} = \frac{2\pi\cdot\mathrm{Z}\cdot\mathrm{e}^2\cdot\mathrm{K}}{\mathrm{n}\cdot\mathrm{h}}$$

$$\therefore \frac{2\pi \mathrm{Ze}^2\cdot\mathrm{K}}{\mathrm{n}\cdot\mathrm{h}} = \frac{\mathrm{n}^2\,\mathrm{h}^2}{4\pi^2 \mathrm{me}^2 \mathrm{Z}\cdot\mathrm{K}} \times \omega;$$

$$\omega = \frac{8\pi^3 \mathrm{me}^4\cdot\mathrm{Z}^2\cdot\mathrm{K}^2}{\mathrm{n}^3\cdot\mathrm{h}^3}$$

$$= \frac{9\pi^3 \mathrm{me}^4\cdot\mathrm{K}^2}{\mathrm{h}^3} (\because \mathrm{n} = 2 \text{ and } \mathrm{Z} = 3)$$

@IITJEE_Advanced

Q.2 Answer:

-30.6eV

Solution:

For Li^{2+} ion

$$E = -13.6 \times \frac{Z^2}{n^2} = -13.6 \times \frac{(3)^2}{(2)^2}$$
$$= \frac{-13.6 \times 9}{4} = -30.6\text{eV}$$

Q.3 Answer:

10.19eV

Solution:

The smallest value of energy of an electron in H atom in ground state can absorb is $= \text{E}_2 - \text{E}_1$
$= \frac{-13.58}{4} - \left(\frac{-13.58}{1}\right) = 10.19$

Q.4 Answer:

$-\frac{1}{2}\frac{ke^2}{r}$

Solution:

Total energy $= \text{K.E.} + \text{P.E.}$
$\text{KE.} = -\frac{1}{2}\text{mv}^2 = \frac{1}{2}\frac{\text{Zke}^2}{\text{r}}$ **(centripetal force)**
P.E. $= -\frac{\text{Zke}^2}{\text{r}}$
T.E. $= \frac{1}{2}\frac{\text{Zke}^2}{\text{r}} + -\frac{\text{Zke}^2}{\text{r}}$
$= -\frac{1}{2}\frac{\text{Zke}^2}{\text{r}}$
$= -\frac{1}{2}\frac{\text{ke}^2}{\text{r}}$

Q.5 Answer:

$1:4$

Solution:

From the expression of Bohr's theory, we know that
$m_e \text{v}_1 r_1 = n_1 \frac{h}{2\pi}$
$\& m_e \text{v}_2 r_2 = n_2 \frac{h}{2\pi}$
$\frac{m_e \text{v}_1 r_1}{m_e \text{v}_2 r_2} = \frac{n_1}{n_2}\frac{h}{2\pi} \times \frac{2\pi}{h}$
Given, $r_1 = 5r_2, n_1 = 5, n_2 = 4$
$\frac{m_e \times \text{v}_1 \times 5r_2}{m_e \times \text{v}_2 \times r_2} = \frac{5}{4}$
$\Rightarrow \quad \frac{\text{v}_1}{\text{v}_2} = \frac{5}{4\times 5} = \frac{1}{4} = 1:4$

Q.6 Answer:

1.51eV

@IITJEE_Advanced

Solution:

2nd **excited state will be the** 3rd **energy level**
$\mathrm{E_n} = \frac{13.6}{\mathrm{n^2}}\mathrm{eV}$ **or** $\mathrm{E} = \frac{13.6}{9}\mathrm{eV} = 1.51\mathrm{eV}$

Q.7 Answer:

$11.58 \times 10^{-44}\ \mathrm{kg\ m^2\ s^{-1}}$

Solution:

$\mathrm{Z} = 3$ **for** Li^{2+} **ions**
So $\mathrm{r}_n = \frac{52.9 \times n^2}{Z}\mathrm{pm}$
$n = 3, Z = 3$
$\mathrm{r}_n = \frac{52.9 \times (3)^2}{3}\mathrm{pm}$
$= 158.7\mathrm{pm}$
Also, linear momentum $(\mathrm{mv}) = 7.3 \times 10^{-34}\ \mathrm{kgms^{-1}}$
Then angular momentum will be $\omega = (\mathrm{mv}) \times \mathrm{r}$
$= \left(7.3 \times 10^{-34}\ \mathrm{kg\ ms^{-1}}\right)(158.7\mathrm{pm})$
$= 7.3 \times 10^{-34}\ \mathrm{kg\ ms^{-1}} \times \left(158.7 \times 10^{-12}\ \mathrm{m}\right)$
$= 11.58 \times 10^{-44}\ \mathrm{kg\ m^2\ s^{-1}}$

Q.8 Answer:

$+\frac{3}{2}x$

Solution:

Change in P.E. $= -\frac{2x}{4} + (2x) = \frac{3}{2}x$

Q.9 Answer:

16

Solution:

$\mathrm{r_1} = 0.529\mathring{A} \quad \mathrm{R_{4(x)}} = \mathrm{R_1} \times \frac{\mathrm{n^2}}{\mathrm{Z}}$
$\mathrm{R_{4(x)}} \Rightarrow \frac{0.529 \times (4)^2}{\mathrm{Z}}; \quad \mathrm{Z} = 16$

Q.10 Answer:

$1 : 32$

Solution:

$\mathrm{T} \propto \frac{\mathrm{n^3}}{\mathrm{Z^2}}; \quad \frac{\mathrm{T_{1,2}}}{\mathrm{T_{2.1}}} = \frac{1}{4} \times \frac{1}{8} = \frac{1}{32}$

Q.11 Answer:

$-217.9 \times 10^{-20}\ \mathrm{J}$

Solution:

@IITJEE_Advanced

$E_1He^+ = E_1H \times Z^2$
$\therefore -871.6 \times 10^{-20} = E_{1H} \times 4$
$\therefore E_{1_H} = -217.9 \times 10^{-20}$ J

Q.12 Answer:

2.25R

Solution:

$\frac{r_1}{r_2} = \left(\frac{n_1}{n_2}\right)^2 = \frac{4}{9}$

Q.13 Answer:

the spectrum of an atom or ion containing one electron only

Solution:

In Bohr's theory while calculating the energy of electron, the potential energy has been found out by considering only the attraction between the electron and nucleus. If there is another electron in the orbit, the potential energy would change due to electron-electron repulsion. Therefore the Bohr's model is meant for all one-electron systems.

Q.14 Answer:

-4.41×10^{-17} J atom $^{-1}$

Solution:

$E = \frac{Z^2}{n^2} \times 13.6\text{eV}$
or $\frac{I_1}{I_2} = \frac{Z_1^2}{n_1^2} \times \frac{n_2^2}{Z_2^2}$
Given $I_1 = -19.6 \times 10^{-18}, Z_1 = 2$
$n_1 = 1, Z_2 = 3$ **and** $n_2 = 1$
Substituting these values in equation (ii). $\frac{19.6 \times 10^{-18}}{I_2} = \frac{4}{1} \times \frac{1}{9}$
$I_2 = -19.6 \times 10^{-18} \times \frac{9}{4}$
$= -4.41 \times 10^{-17}$ J/ **atom**

Q.15 Answer:

$\frac{h^2}{32\pi^2 m a_0^2}$

Solution:

As per Bohr s postulate, $mvr = \frac{nh}{2\pi}$
So, $v = \frac{nh}{2\pi mr}$
$\text{KE} = \frac{1}{2}mv^2$
So, $\text{KE} = \frac{1}{2}m\left(\frac{nh}{2\pi mr}\right)^2$
Since, $r = \frac{a_o \times n^2}{z}$

@IITJEE_Advanced

So, for 2nd Bohr orbit

$r = \frac{a_o \times 2^2}{1} = 4a_o$

$KE = \frac{1}{2}m\left(\frac{2^2h^2}{4\pi^2m^2 \times (4a_o)^2}\right)$

$KE = \frac{h^2}{32\pi^2 m a_o^2}$

Q.16 Answer:

$-\frac{3e^2}{4\Pi \in_0 r^2}$

Solution:

The atomic number of of Li is 3. So, the number of eletcrons present in it is also 3.

Li^{+2} ion is formed by the loss of two electrons.

$\therefore$ The number of electrons in Li^{+2} ion = 3 - 2 = 1.

Now, the number of electrons in Li^{+2} is same as the number of electrons in a hydrogen atom, which also contains only one electron. Therefore, the Bohr's formula for the potential energy of an electron in H-atom can be applied to Li^{2+} ion as well.

From the Bohr's model, the potential energy of an electron in an H-like ion is given as :

$P.E. = \frac{-Ze^2}{4\Pi \in_0 r^2}$

where, Z is the atomic number of the H-like species.

For Li^{2+} ion, Z = 3

Therefore, $P.E. = \frac{-3e^2}{4\Pi \in_0 r^2}$

Q.17 Answer:

-1

Solution:

The ratio $KE/E = -1$

Q.18 Answer:

$1.09 \times 10^6 \text{ m s}^{-1}$

Solution:

Since $v \propto Z/n$, we will have $\frac{v(Li^{2+})}{v(H)} = \frac{3/2}{1/1}$

$\Rightarrow v(Li^+) = \left(\frac{3}{2}\right)v(H) = \left(\frac{3}{2}\right)(7.27 \times 10^5 \text{ m s}^{-1}) = 1.09 \times 10^6 \text{ m s}^{-1}$

@IITJEE_Advanced

Q.19 Answer:

Kinetic energy of an electron in an orbit is equal to the magnitude of its potential energy.

Solution:

Let an electron revolves with a constant speed v around the nucleus in a stationary orbit of radius r.

The electrostatic force acting on the electron is balanced by the centrifugal force i.e. $F_c = F_e$

$\therefore \frac{mv^2}{r} = \frac{ke^2}{r^2}$ **where** $k = \frac{1}{4\pi\epsilon_o}$

or $mv^2 = \frac{ke^2}{r}$ **... (1)**

Thus kinetic energy of electron $K = \frac{1}{2}mv^2 = \frac{ke^2}{2r}$ **... (2)**

Potential energy of electron $U = \frac{-ke^2}{r}$

$\Rightarrow U = -2K$ **(using (2))**

Q.20 Answer:

Zeeman effect

Solution:

Zeeman effect

@IITJEE_Advanced